

Transforming Pharma:

The promise of AI in developing better medicines faster

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Thursday Oct 29th 2026
9 pm EDT

iit ALUMNI

Talk Outline

The process by which a therapeutic idea eventually becomes a drug

What are the stages and gates that a successful drug needs to pass through

A deeper dive into each stage and where AI is beginning to play a role

Limitations

What does the near future look like?

Some Caveats

The talk is focused on the R&D behind getting a drug to market

I have not considered impact on scale-up, manufacturing, commercialization, etc.

The talk is not meant to be all-encompassing, it reflects my opinions

How is a drug discovered?

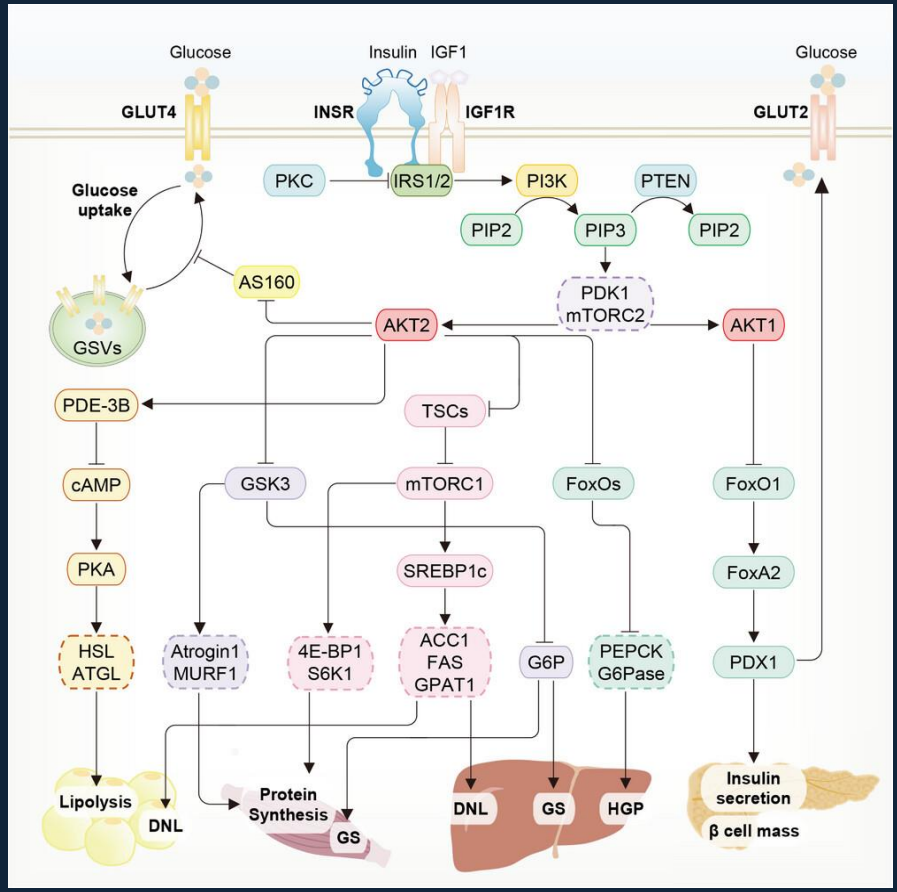
A complex process that starts by **relating a part of your biology to disease symptoms** (target)

Example of Diabetes

Muscles don’t take up (and burn) enough glucose

Transporter that moves glucose from the blood into the muscle is faulty

-> Increase insulin or make the glucose transporter more sensitive



https://doi.org/10.1002/mco2.283

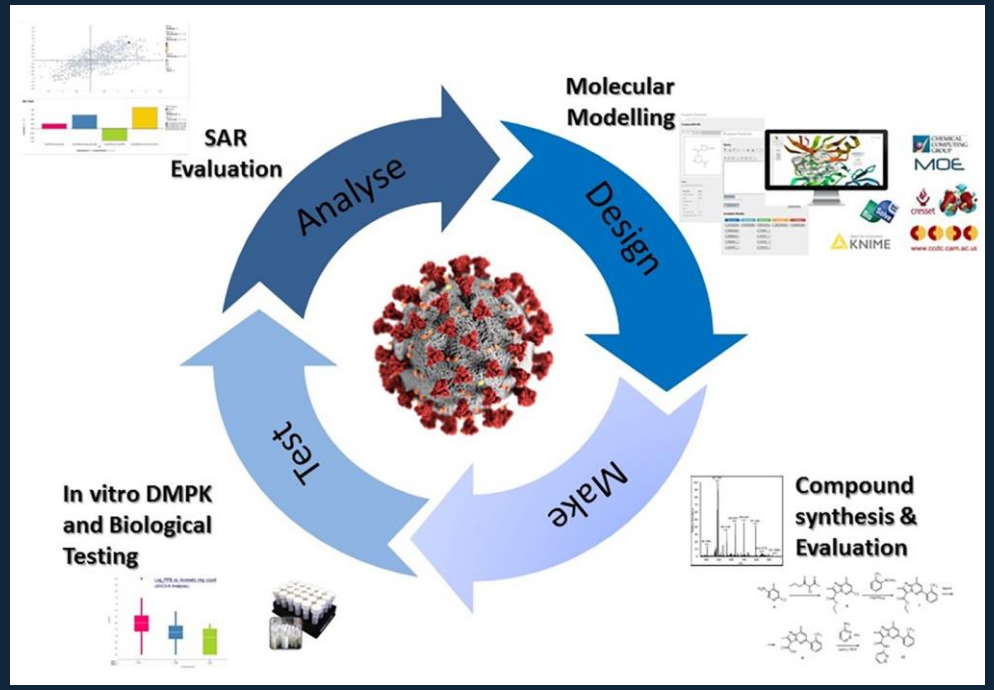
How is a drug discovered?

Design a chemical that makes the target work efficiently

Tested in on cells in a dish (*in vitro*) and in animals (*in vivo*)

Chemical structure is constantly tweaked to optimize its performance

Tested in long-term animal studies to show that it doesn't cause other problems



Adapted from Heifetz A.; *et al*. 2018

How is a drug discovered

Make the chemical needs to behave like a “drug”

it should not interfere with any other biology (it should be safe!)

It should stay in the body for as long as it is needed

It should be stable, you should be able to make it into a pill, it should be manufacturable, etc.



https://www.wuxibiologics.com/in-vivo-pharmacology-services/toxicology/

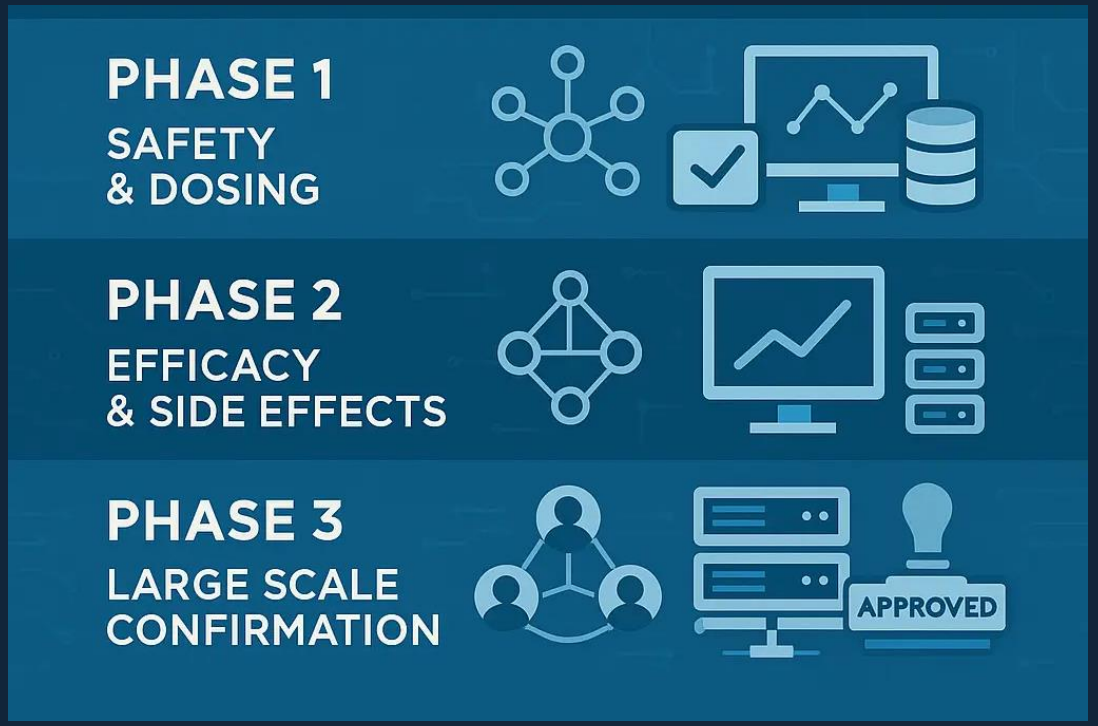
How is a drug discovered

Test in Humans

In small groups of healthy people to show that it is safe (Phase I)

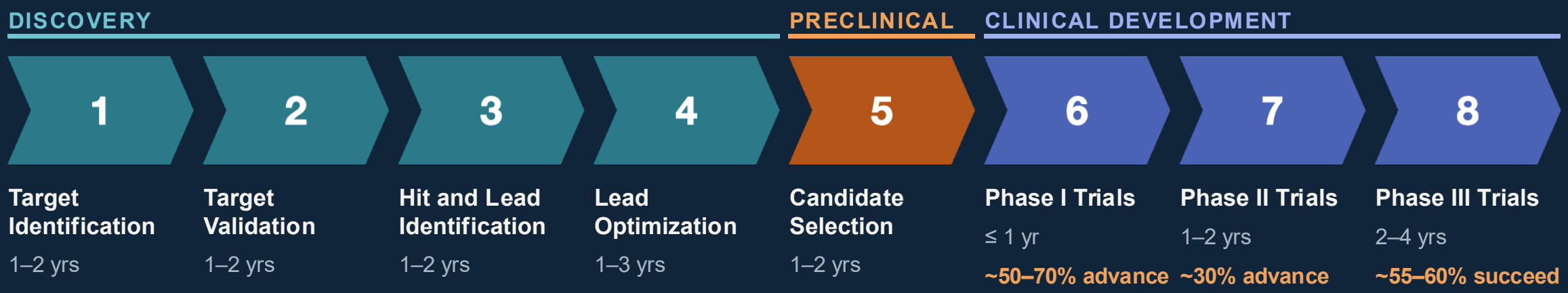
In a small group of patients to figure out that it really works in humans (Phase II)

In large groups of patients (1000s) in different geographies, ethnicities, etc. to confirm it works and it is safe (Phase III)



https://intuitionlabs.ai/articles/clinical-trial-phases-1-3-comprehensive-guide-pharma-it

The Drug Discovery and Development Pipeline



Eight stages from a biological target to a registered medicine, typically 8-10 years

Total Cost of Developing a successful medicine: ~1B USD

Drug Discovery and Development is not Efficient

Process is largely linear through the pipeline

Use of simulation is low – often experiments are exploratory

A lot of “brute force” experimentation

Most of the *in vitro* and *in vivo* systems are poor representations of a human

Plenty of scope for improvement, potentially by AI



Target Identification

Find a gene, protein or pathway that drives a disease and whose function could be changed by a drug

Approaches

- Human genetics and GWAS
- Transcriptomics and proteomics
- CRISPR and functional screens

Questions asked

- Is there an unmet medical need?
- Is it linked to disease biology?
- Is it druggable?

Output

- A ranked list of candidate targets
- Typical time: about 1–2 years

AI IN STAGE 1 OF 8

AI in Target Identification

How AI helps

- Mine complex data sets and literature to discover candidate targets
- Knowledge graphs link genes, diseases and existing drugs
- Flag approved drugs that could be repurposed

Real-world examples

Insilico Medicine: its PandaOmics platform identified TNIK as a novel fibrosis target, which led to the drug rentosertib.

BenevolentAI: in 2020 its knowledge graph flagged baricitinib, an approved arthritis drug, as a possible COVID-19 treatment. Clinical trials later supported its use.

https://www.biopharmatrend.com/post/1282-ai-designed-tnik-inhibitor-shows-lung-function-gains-in-ipf
https://pmc.ncbi.nlm.nih.gov/articles/PMC7137985



Target Validation

Show that changing the target changes the disease safely

Evidence

- Knockdown or knockout (siRNA, CRISPR) in animals
- Tool compounds and antibodies
- Cell and animal disease models

Also checked

- Tissue expression and safety risks
- Biomarkers of target activity
- Assay and druggability feasibility

Output

- A confident go or no-go on the target
- Typical time: about 1–2 years

AI IN STAGE 2 OF 8

AI in Target Validation

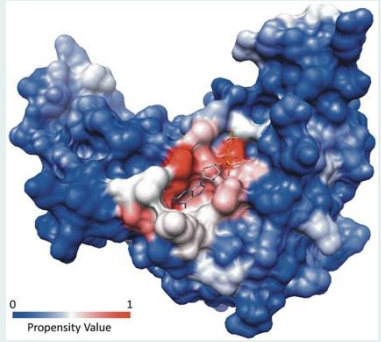
How AI helps

- Mine scientific literature to verify target-disease linkage
- Predict protein structures to judge druggability
- Predict how genetic variants change protein function
- Read cell images to see what a gene knockout does

Real-world examples

AlphaFold database: DeepMind and EMBL-EBI released predicted structures for over 220 million proteins, free to use, so teams can assess binding pockets without waiting for crystal structures.

Recursion: runs automated imaging of cells after gene knockouts and compound treatments, then uses ML to map biological relationships.





Hit and Lead Identification

“Screen” for molecules that bind the target, then turn confirmed hits into a lead series.

Finding hits

- Fragment and DNA-encoded libraries
- Virtual screening and AI design
- High-throughput screening of 100,000+ compounds

Confirming

- Dose–response and potency
- Counter-screens for selectivity
- Early ADME and chemical tractability

Output

- One or more lead series
- Typical time: about 1–2 years

AI IN STAGE 3 OF 8

AI in Hit and Lead Identification

How AI helps

- Virtually screen very large compound libraries
- Generative models propose novel molecular scaffolds
- Predict binding and early ADME to prioritize hits

Real-world examples

Halicin (MIT, 2020): a deep-learning model screened a drug library and found halicin, a structurally new antibiotic active against resistant bacteria.

Abaucin (MIT and McMaster, 2023): the same approach found a compound that targets Acinetobacter baumannii, a dangerous hospital pathogen.



Lead Optimization

Refine the chemistry iteratively to balance potency, selectivity, pharmacokinetics and safety.

Design cycle

- Design, make, test, analyze
- Structure-based and computational design
- Structure–activity relationships

Properties tuned

- Potency and selectivity
- Solubility, permeability, stability
- Toxicity liabilities (hERG, CYP)

Output

- Optimized compounds with in vivo efficacy
- Typical time: about 1–3 years

AI IN STAGE 4 OF 8

AI in Lead Optimization

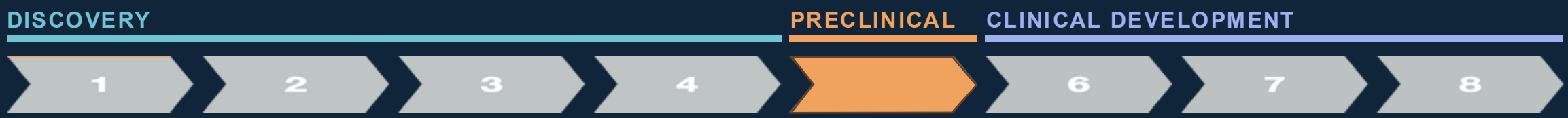
How AI helps

- Predict potency, selectivity and ADMET before synthesis
- Generative design inside design-make-test loops
- Physics-based free-energy calculations combined with ML

Real-world examples

Zasocitinib (Takeda): Nimbus Therapeutics designed this TYK2 inhibitor with Schrödinger's computational platform, then Takeda acquired it.

Phase 3 trials met all primary endpoints, and in September 2026 the FDA accepted its application under Priority Review, with a decision expected in Q1 2027.



Candidate Selection

Choose the one molecule, plus a backup, to take into humans, supported by IND-enabling studies.

Preclinical package

- GLP toxicology in two species
- Safety pharmacology
- Pharmacokinetics and dose prediction

Development readiness

- Formulation and manufacturing scale-up
- Stability and purity (CMC)
- Starting dose rationale

Output

- IND or CTA filing to start trials
- Typical time: about 1–2 years

AI IN STAGE 5 OF 8

AI in Candidate Selection

How AI helps

- Predict toxicity and ADMET liabilities before animal studies
- Estimate human dose from preclinical data
- Use computational models to reduce animal testing

Real-world examples

Rentosertib (Insilico): the program went from target identification to preclinical candidate nomination in about 18 months, according to the company.

FDA roadmap (April 2025): the agency announced plans to reduce animal testing, starting with monoclonal antibodies, using AI-based toxicity models and human-cell systems.



Phase I Trials

First-in-human studies that ask whether the drug is safe and how the body handles it.

Participants

- About 20–100 people
- Mostly healthy volunteers
- Patients in some areas, such as oncology

Studies

- Single and multiple ascending doses
- Safety and tolerability
- Pharmacokinetics and early PD

Outcome

- Maximum Tolerated Dose
- Roughly 50–70% advance
- Duration: months to about 1 year

AI IN STAGE 6 OF 8

AI in Phase I Trials

How AI helps

- Simulate pharmacokinetics to choose starting and escalating doses
- Predict safety signals from preclinical and omics data
- Analyze biomarker and wearable data from volunteers

Real-world example

DSP-1181 (Exscientia and Sumitomo): in 2020 it became one of the first AI-designed molecules to enter Phase I, for OCD. It was later discontinued, so AI can speed entry without ensuring success.



Phase II Trials

Test whether the drug works in patients with the disease and find the right dose.

Participants

- About 100–300 patients
- Often randomized, sometimes placebo-controlled
- IIa proof of concept, IIb dose-finding

Studies

- Efficacy signals and biomarkers
- Dose–response relationship
- Side effects in patients

Outcome

- Go or no-go for Phase III
- Highest attrition: roughly 70% fail
- Duration: about 1–2 years

AI IN STAGE 7 OF 8

AI in Phase II Trials

How AI helps

- Select patients most likely to respond using biomarkers
- Digital twins shrink the placebo arm
- Analyze early response data for adaptive designs

Real-world examples

Unlearn: The EMA qualified Unlearn's method of using digital twins to reduce the size of the placebo arm.

Differentia: in 2025 the EMA used our digital twin simulations to reduce and replace patients for a pulmonary hypertension drug



Phase III Trials

Large confirmatory trials that prove benefit and safety well enough to support approval.

Participants

- About 300 to 3,000+ patients
- Multicenter, randomized, controlled
- Placebo or standard-of-care comparator

Studies

- Primary clinical endpoints
- Long-term safety and subgroups
- Often two pivotal trials

Outcome

- NDA or BLA submission to regulators
- Roughly 55–60% succeed
- Duration: about 2–4 years

AI IN STAGE 8 OF 8

AI in Phase III Trials

How AI helps

- Find eligible patients in health records and choose sites
- Cut sample sizes with digital-twin covariate adjustment
- Monitor data quality and safety signals in real time

Real-world examples

None so far

Limits of AI in Drug Discovery

Data quality and bias

Models learn from existing data, which favors well-studied genes, chemistry and patient populations.

AI generates hypotheses not proof

Targets, hits, toxicity and doses still need lab, animal and clinical confirmation.

Biology is complex, models are flawed

Whole-body effects and patient variability are hard to model, and most candidates still fail in the clinic.
Animals don’t represent the complexity seen in humans

Evidence and oversight

Most AI-discovered drugs are still in trials, many claims come from developers
Regulators don’t like black-box models without explainability

AI speeds up and sharpens decisions; it does not remove clinical risk.

What is the future likely to bring?

Reduction in the timelines

Even incremental improvements in each part of the pipeline will result in overall reductions in time and cost

Less failures, more confirmation

Move the experimental process to confirm hypothesis rather than exploration. All exploration is performed *in silico*

Better, cheaper and productive trials

Use digital twin simulations to select better patients, better doses, better trial designs

Personalized Medicine

Every person has their own digital avatar that can be used to test medicines on, virtually. Make lifestyle changes and take medications the work for YOU, specifically